

Analysis of the two-channel optical Bell test, using multivectors and matrices

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Table of Contents

Introduction

Multivectors and geometric algebra

Vector products and grade

Basis vectors

Rotors

Planes of polarization, vectors, and tensor spaces

Notations

Polarizing beam splitters (PBS)

Correlation

Expectation

Variance and standard deviation

Covariance and correlation

The correlation of the two-channel optical Bell test

Solving two problems at once

A naïve attempt to derive the correlation of f and g

A solution for $\eta = 0$

Realizing that the plane of polarization does not matter

Introduction

I have had it up to my eyeballs with most physicists and their belief in magical “instantaneous action at a distance.”

Yes, I said *magical*.

Some will fault me for calling what appears in “serious” scientific journals (including *Science*, to which I subscribe) as *magic*, but it is nothing less. No physicist should believe in “action at a distance,” and if they do they should be looking for another career. People like Maxwell and Heaviside went to enormous trouble to base physics on action by direct contact. Maxwell’s theory of electromagnetism is thus *explicitly* one of action by direct contact, and the electromagnetic field obeys Newton’s laws of motion. Einstein’s attempt at a theory of gravity, which as far as I know is the best we have so far, is also based *explicitly* on action by direct contact.

Yet we are told by physicists that two or more pulses of light can be “entangled,” thus forming a single physical entity spanning any distance whatsoever. When one of the “entangled” pulses is resolved into a distinct pulse, supposedly the others “*instantaneously*,” over even light years of distance, also resolve into distinct pulses. They do this, even though there is no possibility this is logically consistent with Maxwell’s theory of electromagnetism. There is no possible mathematical resolution to this puzzle, because Maxwell’s theory is one of action by direct contact (expressed as ε - δ limits).

No wonder it is, then, that, when I was taught “Engineering Physics” at the time Alain Aspect was publishing his famous (but in multiple ways incorrect) results, I was also told the following: that “quantum physics” made no logical sense and I should not try to understand it. In fact I was *ordered* in a loud voice not to try to understand it, and *ordered* “Just do the calculations!”

“Quantum physics” does not deserve the title “science.” It is a quasi-religion and a pseudoscience. Dirac notation is a liturgical language for what actually are propositions of probability theory, but physicists transmogrify these propositions, by religious incantations written in Dirac bras and kets, into “entangled” entities.

The “quantum” pseudoscience not only costs university students and their families millions and millions and millions of wasted dollars, but is a threat to world peace. How much does the United States Department of Defense spend on “quantum computers”? Yet “quantum computers” are utter junk that cannot possibly function as physicists predict. We should not tolerate this state of affairs. If every “quantum physicist” should be laughed out of a job, there would be many more competent persons who could approach the same empirical problems using *real* physics and actual mathematics (rather than enchanted liturgy).

A physicist will sometimes brag that the Dirac notation represents a complex tensor or “Hilbert” space, as if this made the mathematics more than just probability theory. In fact, this seems to be what some believe, that the complex tensor space is not merely probability theory, but indeed a magical phase of existence, in which “instantaneous action at a distance” occurs. But such a space is merely a mathematical construct of convenience. It has no physical meaning. I will demonstrate this below by analyzing the two-channel optical Bell test (the Aspect experiment) using not a complex tensor space, but something even fancier. I will use a tensor space of *multivectors*.

Multivectors and geometric algebra

Vector products and grade

Ordinary Gibbs-Heaviside vectors are nice enough, but, like physicists, we want to impress the plebeians. Physicists do this by using complex numbers to represent rotations in a plane. So did I, as a student of electrical engineering, use complex numbers this way. However, are not complex numbers more properly roots of univariate real polynomials? A more direct way to represent rotations is *geometric algebra*, from which we will draw just a miniscule subset of what it contains and can do.

In geometric algebra, you can add a scalar and a vector. At least if you are as old as I am, you are likely to have an acquaintance who will argue with you vociferously that you *cannot* add a scalar and a vector. Such an acquaintance is a person who has obtained a high school or university education, but has not *overcome* that education. It is important to obtain an education, but current educations are very bad, and so it is necessary also to *overcome* the damage done. A school education has solidified the brain rather than made it more plastic, and also much of what is taught is simply wrong, or limited to a context but taught as if it were universal. So wipe that prejudice from your thoughts. We will be adding things together you may have thought you could not add.

You can also multiply vectors in ways other than dot and cross products. The *geometric product* uv of vectors u and v is

$$uv = u \cdot v + u \wedge v$$

where $u \cdot v$ is the familiar dot product, but the *wedge product* $u \wedge v$ is a new entity called a *bivector*. A bivector is a finite planar area that is oriented either counterclockwise or clockwise. You can picture a bivector as an entity normal to which is the familiar cross product.

The sum of any scalars, vectors, and bivectors is called a *multivector*. One can actually go past bivectors to any number of dimensions, and there are other ways things can be made more complicated, but we do not need any of that.

Let us also introduce the *grade* operator, which returns the scalar (*grade 0*), vector (*grade 1*), or bivector (*grade 2*) part of a multivector:

$$\begin{aligned}\langle 5 + 2e_1 + 7e_1 \wedge e_2 \rangle_0 &= 5 \\ \langle 5 + 2e_1 + 7e_1 \wedge e_2 \rangle_1 &= 2e_1 \\ \langle 5 + 2e_1 + 7e_1 \wedge e_2 \rangle_2 &= 7e_1 \wedge e_2\end{aligned}$$

Given the grade operation, the dot and wedge products of two vectors can be obtained from their geometric product:

$$\begin{aligned}u \cdot v &= \langle uv \rangle_0 \\ u \wedge v &= \langle uv \rangle_2\end{aligned}$$

Here are some identities:

$$\begin{array}{ll}u \cdot v = \|u\|\|v\| & \text{if } u \parallel v \\ u \wedge v = 0 & \text{if } u \parallel v \\ u \cdot v = 0 & \text{if } u \perp v \\ u \wedge v = uv = -vu = -v \wedge u & \text{if } u \perp v \\ u \cdot v = v \cdot u & \text{always} \\ u \wedge v = -v \wedge u & \text{always}\end{array}$$

Basis vectors

Suppose we want to have an (x, y) coordinate system for geometric algebra. We can do this by defining *basis vectors* e_1 of magnitude 1 in the x -direction and e_2 of magnitude 1 in the y -direction.

The following are useful identities:

$$\begin{aligned}e_1 \cdot e_2 &= e_2 \cdot e_1 = 0 \\ e_1 \wedge e_1 &= e_2 \wedge e_2 = 0 \\ e_1 \cdot e_1 &= e_1 e_1 = e_2 e_2 = e_2 \cdot e_2 = 1 \\ e_1 \wedge e_2 &= e_1 e_2 = -e_2 e_1 = -e_2 \wedge e_1\end{aligned}$$

Rotors

What if you need a unit vector other than one of the basis vectors? Complex numbers around the clock face are obtained by rotating the number 1: the number 1 is rotated counterclockwise by an angle φ by multiplying it by $e^{i\varphi} = \cos\varphi + i\sin\varphi$. The operation is equivalent to breaking a rightwards-pointing unit vector into components.

There is an analogous operation in geometric algebra. For this operation, one uses a *rotor*. To rotate counterclockwise by φ , the rotor and its inverse are

$$\begin{aligned}R_\varphi &= \cos(\varphi/2) - e_1 e_2 \sin(\varphi/2) \\ R_\varphi^{-1} &= \cos(\varphi/2) + e_1 e_2 \sin(\varphi/2)\end{aligned}$$

Any vector around the clock face can be gotten by rotating e_1 , thus:

$$\begin{aligned}e_{\varphi'} &= R_{\varphi'} e_1 R_{\varphi'}^{-1} = (\cos\varphi' - e_1 e_2 \sin\varphi') e_1 (\cos\varphi' + e_1 e_2 \sin\varphi') \\ &= e_1 \cos^2\varphi' - e_1 e_2 \sin\varphi' \cos\varphi' + \cos\varphi' e_1 e_2 \sin\varphi' - e_1 e_2 \sin^2\varphi' \\ &= e_1 \cos^2\varphi' - e_1 e_2 \sin\varphi' \cos\varphi' + \cos\varphi' e_1 e_2 \sin\varphi' - e_1 e_2 \sin^2\varphi'\end{aligned}$$

where $\varphi' = \varphi/2$. Identities mentioned above allow simplifications:

$$\begin{aligned} -e_1 e_2 e_1 &= e_2 \\ e_1 e_1 e_2 &= e_2 \\ -e_1 e_2 e_1 e_1 e_2 &= -e_1 \end{aligned}$$

Therefore,

$$\begin{aligned} e_{\varphi} &= (\cos^2 \varphi - \sin^2 \varphi) e_1 + (2 \sin \varphi \cos \varphi) e_2 \end{aligned}$$

Let us refer to the *CRC Handbook of Mathematical Sciences, 6th Edition*,^[1] where we find the following double-angle relations on page 170:

$$\begin{aligned} \sin 2\alpha &= 2 \sin \alpha \cos \alpha \\ \cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha \end{aligned}$$

Therefore,

$$e_{\varphi} = e_1 \cos \varphi + e_2 \sin \varphi$$

or, in other words, we can use trigonometry and component addition, just as with complex numbers, except now we are using actual vectors instead of roots of polynomials. Also we have impressed the plebeians with the fancy name “rotor” and by citing from a clinically titled reference book.

Planes of polarization, vectors, and tensor spaces

Notations

The “complex tensor space” of quantum mechanics may seem like magic. Many do indeed seem to believe the space is magical, and that each point in it represents an instance in a phase of existence where “instantaneous action at a distance” occurs. Physicists use funny notations that are not used by mathematicians and therefore can be read as meaning “spooky” (that is, magical) *physical* entities rather than mathematical, purely *verbal* entities.

We instead will use notations similar to those used by mathematicians. My education was in electrical engineering, with some applied mathematics, and my trade was computer programming, so I do not write things as “pure” mathematicians do, and cannot read their abstruse papers. But also, despite electrical engineering being applied physics, I do not take anywhere near the liberties that “pure” physicists do. With the Dirac notation, they have drifted so far from mathematical notations that they do not read the notations as mathematics, but instead as liturgical incantations about physical beings of a mysterious and magical nature. And they do not *reason logically* as they work, but instead *just do the calculations*.

Instead of *bras*, *kets*, and whatnot to represent tensors, all the mathematics below will employ such notations as multivectors and matrices. Thus a pulse of plane-polarized light whose plane of polarization is oriented along the x -axis can be represented by either e_1 or $-e_1$. Let us choose e_1 . Similarly, for a pulse of light plane-polarized normal to the first, the representation can be e_2 . Now suppose “There is a light source that transmits a light pulse e_1 to the left arm of the apparatus and e_2 to the right arm.” This initial *proposition* can be written as a matrix thus:

$$\text{prop}_1 = \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}$$

The upper row of such a matrix represents the left arm of the apparatus, and the lower row represents the right arm. The expression prop_1 defines a point in a tensor space. Its meaning is *not* a magical “entangled pair” of light pulses.^[2] If

we had written the same thing in Dirac notation, that is what it *would have* meant, by “quantum physics” doctrine. But we wrote the same thing in a different notation that comes with no such doctrine, and thus it does not mean that. Instead it means a proposition.

That a point in a space might represent something *verbal* rather than physical may seem strange to many physicists, and difficult to handle. If so, that is their problem, not ours. Let them get other jobs. They are costing university students a fortune for lessons in nonsense, and they are ruining the electronics profession and threatening global peace, with “quantum electronics” and “quantum computers” junk that cannot possibly work as advertised. A “quantum computer” is an expensive and unreliable `WAIT A WHILE THEN PRESS ENTER` device.

Later we will encounter matrices with multiple columns, representing propositions in tensor spaces of higher dimensionality, and weighted sums of matrices, representing propositions of probability theory.

Polarizing beam splitters (PBS)

The effect of a polarizing filter on plane-polarized light is described by the *Law of Malus*: the intensity of the retransmitted light is reduced by a factor of $\cos^2(\vartheta - \varphi)$, where ϑ is the angle of the filter’s axis of transmission and φ is the angle of polarization of the impinging light. Let us suppose the mechanism of light is such that a pulse either is or is not retransmitted. In that case intensity can be equated with probability of retransmission of a light pulse.

Now let us proceed from simple polarizing filters to *polarizing beam splitters (PBS)*, also known as *beam-splitting polarizers*. These are devices made, one way or another, from two complementary polarizing filters, which split impinging light into complementary polarizations. In one output channel the light will be polarized along the main axis of transmission, and in the other channel along the complementary axis, which is normal to the main axis. Ideally, the intensity of light in the complementary channel is reduced by the factor $\sin^2(\vartheta - \varphi)$, so the total intensity is not lost. Similarly, for our probabilistic interpretation, we will make an idealistic assumption: that the light pulse is retransmitted in either the main or the complementary channel, but never lost. The probability of retransmission is $\cos^2(\vartheta - \varphi) + \sin^2(\vartheta - \varphi) = 1$.

Sometimes I visualize the action of a polarizing filter as the projection of the vector of the impinging light onto the axis of polarization of the filter. This gives a vector of magnitude reduced from unity, and the square of that magnitude is the intensity of the retransmitted light. If e_ϑ is a unit vector representing the axis of transmission of the filter, and e_φ the angle of polarization of the impinging light, then the projection is

$$\text{proj}_{e_\vartheta} e_\varphi = (e_\varphi \cdot e_\vartheta) e_\vartheta$$

The probability of retransmission is, conveniently, the geometric product of this projection with itself:

$$(\text{proj}_{e_\vartheta} e_\varphi)^2 = (e_\varphi \cdot e_\vartheta)^2 e_\vartheta e_\vartheta = \cos^2(\vartheta - \varphi)$$

The initial light pulses are plane-polarized e_1 and e_2 , and in a PBS there are two polarizers: the main channel e_ϑ and the complementary channel $e_{\vartheta+\pi/2}$. Therefore there are the following special instances:

$$\begin{aligned} (\text{proj}_{e_\vartheta} e_1)^2 &= (e_1 \cdot e_\vartheta)^2 e_\vartheta^2 = \cos^2 \vartheta \\ (\text{proj}_{e_\vartheta} e_2)^2 &= (e_2 \cdot e_\vartheta)^2 e_\vartheta^2 = \sin^2 \vartheta \\ (\text{proj}_{e_{\vartheta+\pi/2}} e_1)^2 &= (e_1 \cdot e_{\vartheta+\pi/2})^2 e_{\vartheta+\pi/2}^2 = \cos^2(\vartheta + \pi/2) = \sin^2 \vartheta \\ (\text{proj}_{e_{\vartheta+\pi/2}} e_2)^2 &= (e_2 \cdot e_{\vartheta+\pi/2})^2 e_{\vartheta+\pi/2}^2 = \sin^2(\vartheta + \pi/2) = \cos^2 \vartheta \end{aligned}$$

The apparatus of the experiment has two PBS. Let us call the angles of the main axes of transmission ζ (zeta) for the left PBS and η (eta) for the right PBS. For the complementary channels, the angles are $\zeta + \pi/2$ and $\eta + \pi/2$. Thus on the left side the retransmitted light pulse can be described either as e_ζ or $e_{\zeta+\pi/2}$, depending on which channel was taken.

Similarly, on the right side, the retransmitted light pulse is e_η or $e_{\eta+\pi/2}$.

Thus the facts of the experiment, taking into account *everything* stated so far, can be written as a point in a tensor space. In the following proposition, P_1, P_2, P_3, P_4 represent mutual probabilities of the possibilities represented by the matrix columns:

$$\begin{aligned} \text{prop}_2 = & P_1 \begin{pmatrix} e_\zeta & e_1 \\ e_\eta & e_2 \end{pmatrix} + P_2 \begin{pmatrix} e_{\zeta+\pi/2} & e_1 \\ e_\eta & e_2 \end{pmatrix} \\ & + P_3 \begin{pmatrix} e_\zeta & e_1 \\ e_{\eta+\pi/2} & e_2 \end{pmatrix} + P_4 \begin{pmatrix} e_{\zeta+\pi/2} & e_1 \\ e_{\eta+\pi/2} & e_2 \end{pmatrix} \end{aligned}$$

Again the upper rows represent the left side of the apparatus and the lower rows represent the right side.

(A tongue-twisting exercise for the reader: Can you rephrase prop_2 in ordinary language?)

The probability of $\text{prop}_1 = \begin{pmatrix} e_1 & e_2 \end{pmatrix}^t$ is one, and thus each of P_1, P_2, P_3, P_4 equals the conditional probability of the left column, given the facts described by the right column (prop_1). This is true by the definition of the conditional probability,

$$P(\text{prop}_x | \text{prop}_y) = \frac{P(\text{prop}_x \wedge \text{prop}_y)}{P(\text{prop}_y)}$$

where in this case $P(\text{prop}_y) = P(\text{prop}_1) = 1$.

Although John S. Bell famously interpreted conditional probability to mean “no action at a distance,”^[3] and used this interpretation to elide facts from the EPR-B experiment (whereas we made certain to include *every fact*), conditional probability *has no such meaning*. The *mathematical definition* of conditional probability is *exactly* what is written above. You can look this up in a text on probability theory. You can also verify, by playing with collections of ordinary objects, that conditional probability as defined above has *the following* intuitive meaning (rather than Bell’s): it is the probability prop_x is true, given that prop_y is true.

This means we calculate P_1, P_2, P_3, P_4 by assuming e_1 on the left and e_2 on the right. We would have done so anyway, but it is nice to see the mathematics even subtly agreeing with us.

Nothing in the formalism implies, as Bell assumes it does, anything about detections in opposite arms of the experiment affecting each other.

A tentative formula for computing the probabilities is as follows. If

$$M_n = \begin{pmatrix} e_{\angle a} & e_1 \\ e_{\angle b} & e_2 \end{pmatrix}$$

then

$$P_n = \overset{?}{(\text{proj}_{e_{\angle a}} e_1)^2 (\text{proj}_{e_{\angle b}} e_2)^2}$$

which is a mutual probability of the top row and bottom row. We do not have to do probability theory calculations to eliminate e_1 and e_2 , because the projection operations already have eliminated these vectors. Furthermore, the

probability of one polarizer setting would seem not to be conditional on the probability of the other polarizer setting, because both these settings are arbitrary. Thus computing P_n might simply be multiplying the probabilities of the two rows.

Or is it? We will see later on it is not so simple. In probability theory, where there seems to be an impediment, there probably is an impediment.

Now prop_2 can be written out more fully, but with doubt that what we are writing is correct:

$$\begin{aligned} \text{prop}_2 = & P_1 \begin{pmatrix} e_\zeta & e_1 \\ e_\eta & e_2 \end{pmatrix} + P_2 \begin{pmatrix} e_{\zeta+\pi/2} & e_1 \\ e_\eta & e_2 \end{pmatrix} \\ & + P_3 \begin{pmatrix} e_\zeta & e_1 \\ e_{\eta+\pi/2} & e_2 \end{pmatrix} + P_4 \begin{pmatrix} e_{\zeta+\pi/2} & e_1 \\ e_{\eta+\pi/2} & e_2 \end{pmatrix} \end{aligned}$$

where

$$\begin{aligned} P_1 &= \cos^2 \zeta \sin^2 \eta \\ P_2 &= \sin^2 \zeta \sin^2 \eta \\ P_3 &= \cos^2 \zeta \cos^2 \eta \\ P_4 &= \sin^2 \zeta \cos^2 \eta \end{aligned}$$

The tensor prop_2 is a proposition stating the probability of each possible event in the two-channel Bell test experiment, in terms of ζ and η . Be very careful, though, in how you evaluate that! Those angles are defined relative to e_1 . But e_1 itself points in some unstated direction. So how would you know whether the PBS were oriented correctly?

We do, however, have a result that gives a probability for every possible event in the experiment and which can be checked for summation to one:

$$\begin{aligned} P_1 + P_2 + P_3 + P_4 &= \cos^2 \zeta \sin^2 \eta + \sin^2 \zeta \sin^2 \eta \\ &\quad + \cos^2 \zeta \cos^2 \eta + \sin^2 \zeta \cos^2 \eta \\ &= (\cos^2 \zeta)(\sin^2 \eta + \cos^2 \eta) \\ &\quad + (\sin^2 \zeta)(\sin^2 \eta + \cos^2 \eta) \\ &= \cos^2 \zeta + \sin^2 \zeta \\ &= 1 \end{aligned}$$

Correlation

Expectation

Let us make a small diversion onto topics of probability theory. The first topic will be *expectation*.

An expectation is an average weighted by probabilities, by frequencies of occurrence, or by some similar fractions that add up to one. For example, suppose there are probabilities P_i that add to one, and associated numbers $f(i)$. Then the expectation of f is

$$E(f) = \sum_i P_i f(i)$$

Variance and standard deviation

Given the expectation of f , it would seem interesting to measure the difference between $f(i)$ and the expectation, but ignoring whether the difference is negative or positive. The sign can be ignored by squaring the difference.

Thus we have the *variance* of f :

$$\begin{aligned}\text{Var}(f) &= \sigma_f^2 = E\left((f - E(f))^2\right) \\ &= \sum_i \left(P_i \left(f(i) - \sum_{\mu} P_{\mu} f(\mu) \right)^2 \right)\end{aligned}$$

The square root of the variance has the same dimensions as f and is called the *standard deviation*:

$$\sigma_f = \sqrt{\text{Var}(f)} = \sqrt{\sigma_f^2}$$

Covariance and correlation

If there are two sets of numbers, $f(i)$ and $g(j)$, it can be useful to compute a figure analogous to the variance, but involving both sets. Such a figure is called the *covariance*:

$$\begin{aligned}\text{cov}_{fg} &= \sigma_{fg} = E((f - E(f))(g - E(g))) \\ &= \sum_{i,j} P_{ij} \left(f(i) - \sum_{\mu} P_{\mu} f(\mu) \right) \left(g(j) - \sum_{\nu} P_{\nu} g(\nu) \right)\end{aligned}$$

A dimensionless figure called the *correlation* is obtained by dividing the covariance of f and g by the respective standard deviations:

$$\text{corr}_{fg} = \rho_{fg} = \frac{\sigma_{fg}}{\sigma_f \sigma_g}$$

The sign of a correlation depends on the definitions of f and g and tends to vary by author.

For the two-channel optical Bell test, there is a dimensionless figure physicists call the “quantum correlation” that equals $\pm \cos 2(\alpha - \beta)$, where α and β are the PBS settings. If physicists are correct, this figure characterizes the supposedly “spooky action at a distance” of the experiment, and should not be derivable without quantum mechanics. Yet below we will derive it as the ordinary correlation, even though we use no quantum mechanics and obviously assume only action by direct contact.^[4]

The correlation of the two-channel optical Bell test

Solving two problems at once

Let

$$\begin{aligned}M_1 &= \begin{pmatrix} e_{\zeta} & e_1 \\ e_{\eta} & e_2 \end{pmatrix} & M_2 &= \begin{pmatrix} e_{\zeta+\pi/2} & e_1 \\ e_{\eta} & e_2 \end{pmatrix} \\ M_3 &= \begin{pmatrix} e_{\zeta} & e_1 \\ e_{\eta+\pi/2} & e_2 \end{pmatrix} & M_4 &= \begin{pmatrix} e_{\zeta+\pi/2} & e_1 \\ e_{\eta+\pi/2} & e_2 \end{pmatrix}\end{aligned}$$

Then

$$\text{prop}_2 = P_1 M_1 + P_2 M_2 + P_3 M_3 + P_4 M_4$$

Define f as follows:

$$\begin{aligned} f(M_1) = f(M_3) &= +1 \\ f(M_2) = f(M_4) &= -1 \end{aligned}$$

Thus f equals $+1$ if the left PBS retransmits in its main channel, -1 if it retransmits in its complementary channel.

The expectation of f is tentatively

$$\begin{aligned} E(f) &= \cos^2 \zeta \sin^2 \eta - \sin^2 \zeta \sin^2 \eta \\ &\quad + \cos^2 \zeta \cos^2 \eta - \sin^2 \zeta \cos^2 \eta \\ &= \cos^2 \zeta - \sin^2 \zeta \end{aligned}$$

which is the probability of a retransmission in the main channel minus the probability of a retransmission in the complementary channel.

(There is reason to believe this is a correct answer, despite that it was derived by questionable means. The reason is that η disappears from the answer.)

We would have liked for the expectation to equal zero. So let us use a trick.^[5] Let us solve two problems simultaneously. Let us solve also for the case where the initial light pulses are $\begin{pmatrix} e_2 & e_1 \end{pmatrix}^t$. A weighted sum of the two light-pulse matrices gives prop_3 that has symmetry:

$$\text{prop}_3 = \frac{1}{2} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} e_2 \\ e_1 \end{pmatrix}$$

Remember that in prop_1 the axis of transmission was not specified. Thus prop_3 is, in effect, the same problem as prop_1 . The new proposition says “The orthogonal planes of polarization have one half probability of being a certain way, one half the opposite way, but the ‘certain way’ is unspecified.” If we analyze that problem, we are for practical purposes analyzing the same problem as the original.

We now need four more matrices:

$$\begin{aligned} M_5 &= \begin{pmatrix} e_\zeta & e_2 \\ e_\eta & e_1 \end{pmatrix} & M_6 &= \begin{pmatrix} e_{\zeta+\pi/2} & e_2 \\ e_\eta & e_1 \end{pmatrix} \\ M_7 &= \begin{pmatrix} e_\zeta & e_2 \\ e_{\eta+\pi/2} & e_1 \end{pmatrix} & M_8 &= \begin{pmatrix} e_{\zeta+\pi/2} & e_2 \\ e_{\eta+\pi/2} & e_1 \end{pmatrix} \end{aligned}$$

And I will leave it as an exercise for the reader to show that the new (but still questionable) probability formula is

$$P_n = \begin{cases} \frac{1}{2} (\text{proj}_{e_{\zeta a}} e_1)^2 (\text{proj}_{e_{\zeta b}} e_2)^2 & \text{if } n \in \{1, 2, 3, 4\} \\ \frac{1}{2} (\text{proj}_{e_{\zeta a}} e_2)^2 (\text{proj}_{e_{\zeta b}} e_1)^2 & \text{if } n \in \{5, 6, 7, 8\} \end{cases}$$

so that

$$P'_1 = P'_8 = (1/2)\cos^2\zeta \sin^2\eta$$

$$P'_2 = P'_7 = (1/2)\sin^2\zeta \sin^2\eta$$

$$P'_3 = P'_6 = (1/2)\cos^2\zeta \cos^2\eta$$

$$P'_4 = P'_5 = (1/2)\sin^2\zeta \cos^2\eta$$

Define

$$f(M_5) = f(M_7) = +1$$

$$f(M_6) = f(M_8) = -1$$

so f still means $+1$ for the main channel. Then

$$\begin{aligned} 2 E(f) = & \cos^2\zeta \sin^2\eta - \sin^2\zeta \sin^2\eta \\ & + \cos^2\zeta \cos^2\eta - \sin^2\zeta \cos^2\eta \\ & + \sin^2\zeta \cos^2\eta - \cos^2\zeta \cos^2\eta \\ & + \sin^2\zeta \sin^2\eta - \cos^2\zeta \sin^2\eta = 0 \end{aligned}$$

A desirable value for the variance, and thus also of the standard deviation, is one. So let us see if we get that. The correlation is designed to eliminate the effects of the particular choices for functions such as f , but that is what lets us choose our f to give us expectation and standard deviation that are easy to work with.

$$\begin{aligned} 2 \sigma_f^2 = & (\cos^2\zeta \sin^2\eta) (+1 - 0)^2 + (\sin^2\zeta \sin^2\eta) (-1 - 0)^2 \\ & + (\cos^2\zeta \cos^2\eta) (+1 - 0)^2 + (\sin^2\zeta \cos^2\eta) (-1 - 0)^2 \\ & + (\sin^2\zeta \cos^2\eta) (+1 - 0)^2 + (\cos^2\zeta \cos^2\eta) (-1 - 0)^2 \\ & + (\sin^2\zeta \sin^2\eta) (+1 - 0)^2 + (\cos^2\zeta \sin^2\eta) (-1 - 0)^2 = 2 \end{aligned}$$

Indeed, then, $\sigma_f = 1$ as desired.

Define a function g that is similar to f , but responds to the right-hand PBS instead of the left:

$$f(M_1) = f(M_3) = f(M_5) = f(M_7) = +1$$

$$f(M_2) = f(M_4) = f(M_6) = f(M_8) = -1$$

$$g(M_1) = g(M_2) = g(M_5) = g(M_6) = +1$$

$$g(M_3) = g(M_4) = g(M_7) = g(M_8) = -1$$

The function g , like f , has expectation zero and standard deviation one.

On either side, which channel has positive sign and which negative is a matter of preference. With some settings the correlation will have a plus sign in front of a “cos” and others a negative sign, and again it makes no real difference. Sometimes you will see it one way, sometimes the other. Our result will have a minus sign.

A naïve attempt to derive the correlation of f and g

Because the expectations of f and g are both zero, and their standard deviations are both one, the covariance and correlation of f and g are equal to each other. In general they would be different at least in that correlation is dimensionless, but our f and g already are dimensionless.

We can thus compute the covariance and treat the result as the correlation.

Let us try to derive the covariance of f and g by the formula

$$\text{cov}_{fg} = \sum_{i,j \in 1, \dots, 8} P_i P_j f(M_i) g(M_j)$$

This results in expressions similar to those of the CHSH inequality.

A little thought makes obvious a tremendous problem with this approach: the mathematics does not distinguish effects that are due to ζ from those that are due to η . Polarizers could be stuck willy-nilly anywhere within the apparatus, and the mathematics cannot tell the difference. *We have made unwarranted assumptions about the mutual probabilities.* We have failed to consider the mutual probabilities might not be products of “independent” probabilities. The resulting expressions are useless.

The CHSH inequality also is useless.

That physicists are *permitted* to be so immensely incompetent in probability theory is scandalous. They teach equal incompetence to students and a next generation, who are fleeced of their possessions for the privilege of being made so stupid. Global peace and the defense of the United States are threatened. No scientific field should be allowed such ignorance.

My education was in signal processing, in the 1980s before electrical engineering was so badly corrupted with “quantum” nonsense. I look for the *actual* solution to a problem such as the Bell test, rather than for some fantasy about entangled fairies and teleporting unicorns. So I work at it until an effective method comes to mind.

What works is to *specialize* the problem so there is only one polarizer setting. Then that specialized solution can be generalized. (This is the reappearance of *specialization* mentioned in [5])

A solution for $\eta = 0$

It $\eta = 0$, then on the right side every e_1 is retransmitted and every e_2 is blocked. Thus the symbol η disappears from expressions. Physically, it is as if we put only a photodetector on that side of the apparatus, and had no polarizing filters on the right at all.

The difficulty with expressions containing both ζ and η exists no longer. Effectively there is no right-side PBS. Now also all effects on the left are from the left-side PBS and are expressed through ζ .

We now have

$$\begin{aligned} P'_1 &= P'_8 = 0 \\ P'_2 &= P'_7 = 0 \\ P'_3 &= P'_6 = (1/2)\cos^2\zeta \\ P'_4 &= P'_5 = (1/2)\sin^2\zeta \end{aligned}$$

and can compute the correlation correctly:

$$\begin{aligned} \rho_{fg} = \text{cov}_{fg} &= (1/2)(\cos^2\zeta)(f(M_3)g(M_3) + f(M_6)g(M_6)) \\ &\quad + (1/2)(\sin^2\zeta)(f(M_4)g(M_4) + f(M_5)g(M_5)) \\ &= -(\cos^2\zeta - \sin^2\zeta) \\ &= -\cos 2\zeta \\ &= -\cos 2(\zeta - \eta) \end{aligned}$$

The next to last step is by one of the double-angle relations earlier quoted from the *CRC Handbook of Mathematical Sciences*. The last step is true because $\eta = 0$.

The result is the same as the “quantum correlation” for the given PBS settings. But there is still one more step before we get the complete answer.

Realizing that the plane of polarization does not matter

That step is to add zero to $\zeta - \eta$.

But not just any zero. Let δ be any angle whatsoever. Also let $\alpha = \zeta + \delta$ and $\beta = \eta + \delta$. Then

$$\begin{aligned} -\cos 2(\alpha - \beta) &= -\cos 2((\zeta + \delta) - (\eta + \delta)) \\ &= -\cos 2((\zeta - \eta) + (\delta - \delta)) \\ &= -\cos 2((\zeta - \eta) + 0) \\ &= -\cos 2(\zeta - \eta) \end{aligned}$$

Thus the PBS settings can be rotated together to any new calibration, without this rotation changing the form of the correlation. However, let us assume there is a standard calibration for PBS. We cannot recalibrate. There remains, however, an equivalent interpretation: that the polarization angles of the initial light pulses do not matter towards the correlation, as long as the two planes of polarization are normal to each other.

This is because the correlation has turned out to be a function of relative angle, expressed as a cosine.

We needed this invariance over in-unison rotation of the PBS settings, anyway. We had no way to know how to set the PBS, otherwise. They must be set only relative to each other, not relative to the light source.

There is a factor of two in $-\cos 2(\alpha - \beta)$ that does not appear if the experiment were to be done with spin-1 particles. Mathematically, the only difference is a change of factor in the independent variables. Physically, I would give polarizing filters the “projection” interpretation described above, whereas for spin-1 particles the device (such as a Stern-Gerlach magnet) is perhaps more like a rectangular window that is open only a certain amount. I base this interpretation on some investigation of trigonometric identities.

In any instance, the goal is achieved. I have derived the correlation of the two-channel optical Bell test, without quantum mechanics and assuming only action by direct contact. The correlation is

$$\rho_{fg} = -\cos 2(\alpha - \beta)$$

where α and β are the respective axes of transmission of main channels of the two PBS.

This is the same expression as the so-called “quantum correlation,” and it is correct, but we can see that “quantum” is a misnomer. It is the ordinary correlation from probability theory. It also can be derived from classical coherence theory of electromagnetic waves.^[6] The only thing mysteriously “quantum” about the expression is how any “quantum” physicist obtained it by *some* means, given the field’s habit of uncritically publishing fallacious semi-solutions such as Bell’s and Clauser’s.

Indeed, given the “quantum correlation” as written above, recognizing it is invariant as long as the difference $\alpha - \beta$ is preserved, and so, setting $\beta = 0$, one could then work backwards and show that no quantum mechanics was necessary for a derivation. That no such “reverse derivation” appears in the “elite journals” is testimony to the incompetence of a physics community that awarded the Nobel prize to Clauser, Aspect, and Zeilinger. Their research should have been withdrawn instead of awarded.

Bell’s *Speakable and Unspeakable in Quantum Mechanics*^[7] ought to be reclassified as “Pathological Science.” It would be one of the most pathological examples.

Let me finish by repeating my harshness is because the defense of the United States is at stake, and because students are being fleeced to be taught nonsense. “Collegial politeness” is unaffordable. In any case I am not a colleague of “professional physicists,” and they would not hesitate to tell me so. I am an amateur with no institutional affiliation, my education is in electrical engineering, and my trade was computer programming. I am expected to be polite to professional physicists, but they are not polite to me. Gone are the days when Benjamin Franklin or Michael Faraday could become Fellow of the Royal Society.

An excuse will be made that “specialization is necessary now and you cannot be a polymath,” and yet here they are, physicists, pretending to be able to do random process analysis without even bothering to study the subject. A reasonable conclusion is they believe they are so smart they need not study the subject. They can intuit it all with their gigantic brains, so much bigger than mine and yours. And that Benjamin Franklin and Michael Faraday were too stupid to do what today’s so very “professional” and specialized physicists do.^[8]

If they really were so smart, they would not believe in entangled fairies and teleporting unicorns.

I finish in this subsection, rather than with a separate formal conclusion, because failure to realize one is free to recalibrate the apparatus is perhaps the worst failure to read the plain meaning of a mathematical expression physicists have ever made. □

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1. W. H. Beyer, ed., *CRC Handbook of Mathematical Sciences*, 6th ed., CRC Press, Boca Raton, FL, 1987.
 2. “Photons,” if we want to use a designation with stronger connotations.
 3. J. S. Bell, *Bertlmann’s socks and the nature of reality*, Preprint CERN-TH-2926, CERN, Geneva, 1980. <https://cds.cern.ch/record/142461/files/198009299.pdf>
 4. I believe you will also find, if you go to the trouble, that violations of Bell or CHSH inequalities are inconsistent with this correlation function, no matter how it be derived. The inequalities are incorrect mathematics. The experimental technique of Alain Aspect and others must be flawed, for it should be impossible to obtain violations of the inequalities.
 5. There is no “physics” in this trick. It is a *mathematical* maneuver to take advantage of symmetry, and I would classify it as an example of George Pólya called the *specialization* heuristic. (G. Pólya, *How to Solve It: A New Aspect of Mathematical Method*, expanded Princeton Science Library ed., Princeton University Press, Princeton, NJ, 2004, With a foreword by John H. Conway.) The specialization heuristic will reappear later.
 6. This was done by A. F. Kracklauer. Of course such coherence calculations are similar to ours, except they are done with continuous probability or measure theory.
 7. J. S. Bell, *Speakable and Unspeakable in Quantum Mechanics*, Cambridge Univ. Press, Cambridge, 1987.
 8. As if either of these Fellows of past centuries would have run to *Nature*, with a paper declaring “faster than light motion” because he had a screw loose. Desiring knowledge rather than a Nobel prize, and knowing the measurements were wrong, they would quietly have overhauled their equipment.